Parametric Clustering

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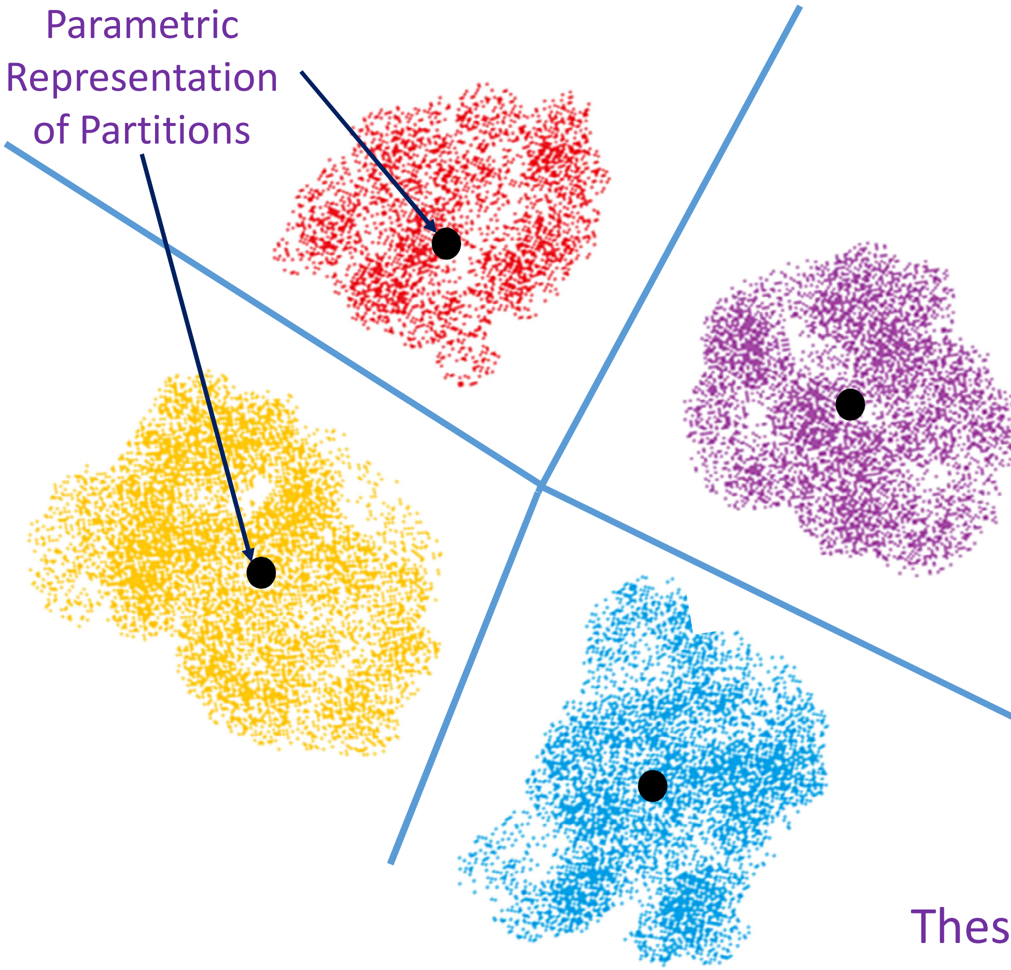
Parametric Clustering

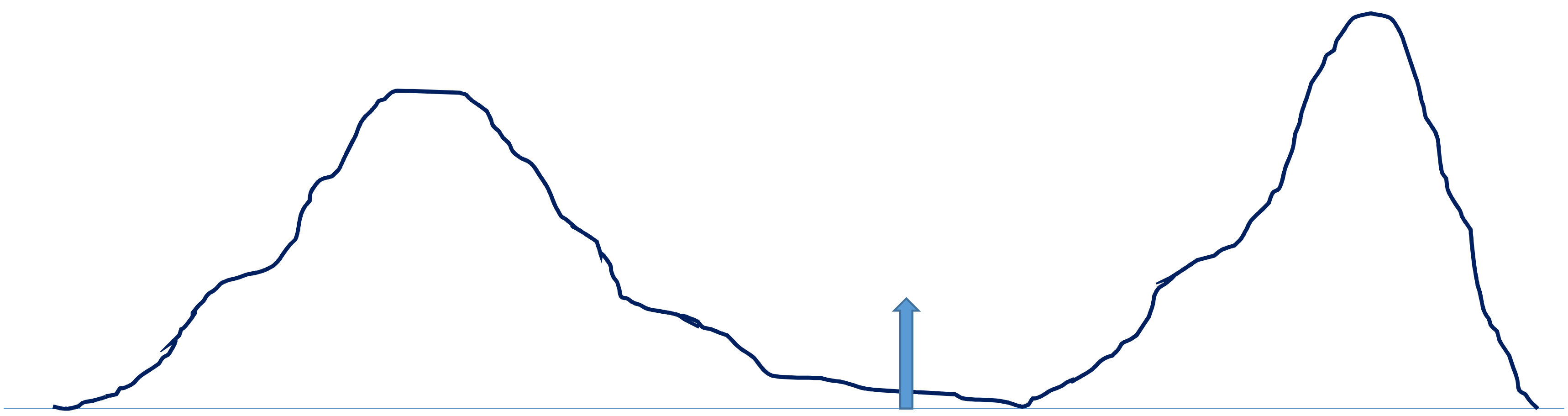


- Adaptive Thresholding
- K-Means
- Hierarchical K-Means

These Algorithms Work on Numerical Input Data

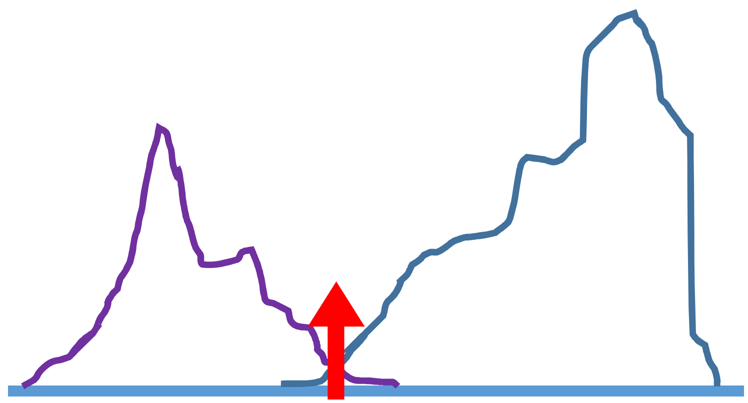
Clustering Algorithms Partition the Input Space into Non-overlapping Regions

Adaptive Thresholding



	Poverty	White	Crime	Doctors	University	Income
Alabama	15.7	71.0	448	218.2	22.0	42,666
Alaska	8.4	70.6	661	228.5	27.3	68,460
Arizona	14.7	86.5	483	209.7	25.1	50,958
Arkansas	17.3	80.8		203.4	18.8	38,815
California	13.3	76.6	523	268.7	29.6	61,021
Colorado		89.7	348	259.7	35.6	56,993
Connecticut	9.3	84.3	256	376.4	35.6	68,595
Delaware	10.0	74.3	689	250.9	27.5	57,989
Florida	13.2	79.8	723	247.9	25.8	47,778
Georgia	14.7	65.4	493	217.4	27.5	50,861
Hawaii	9.1	29.7	273	317.0	29.1	67,214
Idaho		94.6	239	168.8	24.0	47,576
Illinois	12.2	79.1	533	280.2	29.9	56,235
Indiana	13.1			216.9		47,966
Iowa	11.5	94.2	295	189.3	24.3	48,980
Kansas	11.3	88.7	453	222.5	29.6	50,177
Kentucky	17.3	89.9	295	232.3	19.7	41,538
Louisiana	17.3	64.8	730	262.7	20.3	43,733
Maine	12.3	96.4	118	278.4	25.4	46,581

Adaptive Thresholding



Otsu's Method

Adaptive Thresholding: Algorithm

Given $\boldsymbol{D} = \{x_1, \dots x_i, \dots x_n\}$

Initialize Threshold: T_0

Iteration at Step *k* : Threshold T_{k-1} is available

$$\boldsymbol{S}_L^k = \{x: [x < T_{k-1}] \wedge [x \in \boldsymbol{D}]\};$$

$$\boldsymbol{S}_H^k = \{x: [x \geq T_{k-1}] \wedge [x \in \boldsymbol{D}]\};$$

$$\mu_L^k = \frac{\sum_{x \in \boldsymbol{S}_L^k} x}{|\boldsymbol{S}_L^k|}$$

$$\mu_H^k = \frac{\sum_{x \in \boldsymbol{S}_H^k} x}{|\boldsymbol{S}_H^k|}$$

$$T_k = \frac{\mu_L^k + \mu_H^k}{2}$$

Termination: $|T_k - T_{k-1}| \leq \epsilon$

Adaptive Thresholding: Example

$$D = \{-1, 27, 31, 2, 59, 3, 61, 34, 0, 12\}$$

$$T_0 = 3$$

Adaptive Thresholding: Example

$$T_0 = 3 \begin{cases} S_L^1 = \{-1\,, 2\,, 0\} \longrightarrow \mu_L^1 = 0.3333 \\ \\ S_H^1 = \{27\,, 31\,, 59\,, 3\,, 61\,, 34\,, 12\} \longrightarrow \mu_H^1 = 32.4286 \end{cases}$$

$$T_1 = \frac{\mu_L^1 + \mu_H^1}{2} = \frac{0.3333 + 32.4286}{2} = 16.381$$

$$|T_1 - T_0| = 13.381$$

Adaptive Thresholding: Example

$$T_1 = 16.381 \begin{cases} S_L^2 = \{-1, 2, 3, 0, 12\} \Rightarrow \mu_L^2 = 3.2 \\ S_H^2 = \{27, 31, 59, 61, 34\} \Rightarrow \mu_H^2 = 42.4 \end{cases}$$

$$T_2 = \frac{\mu_L^2 + \mu_H^2}{2} = \frac{3.2 + 42.4}{2} = 22.8$$

$$|T_2 - T_1| = 6.419$$

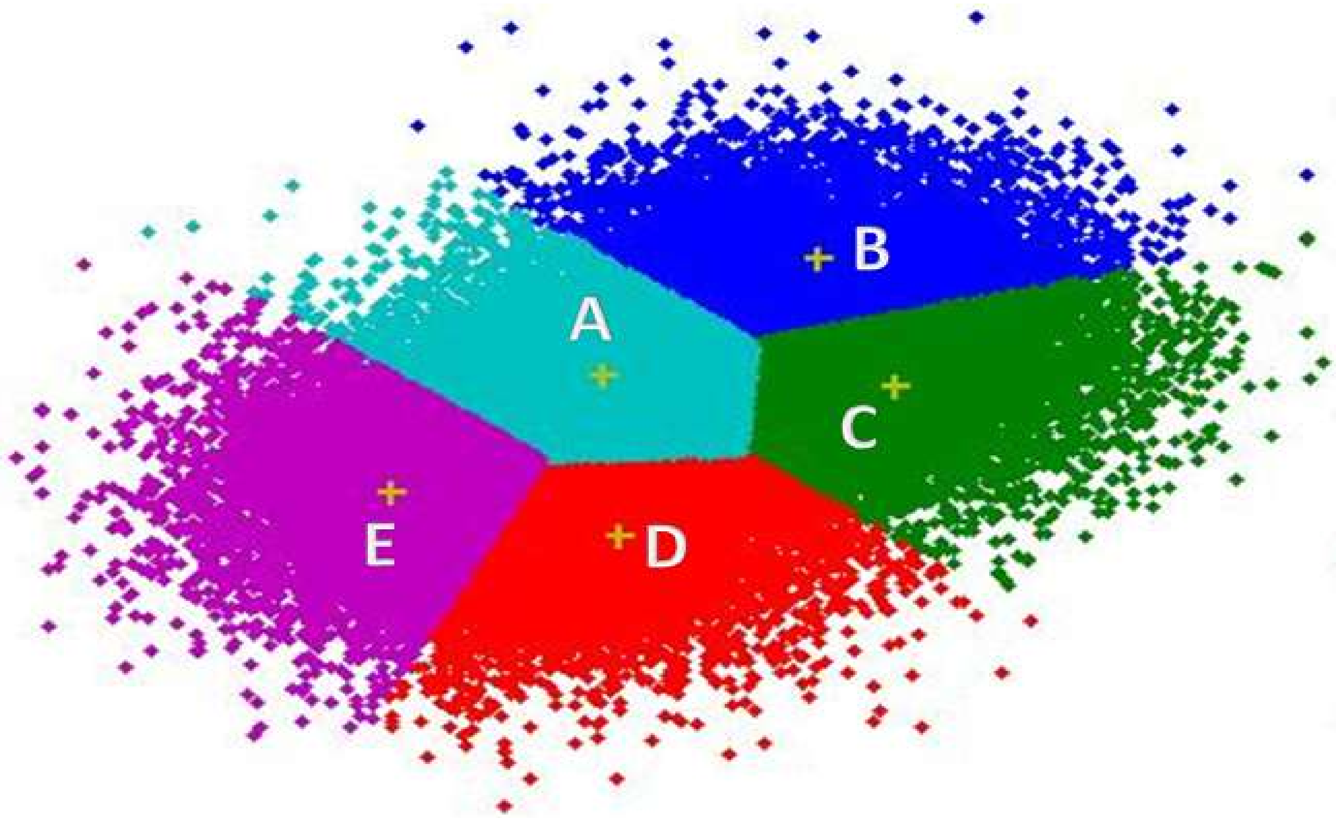
Adaptive Thresholding: Example

$$T_2 = 22.8 \begin{cases} S_L^3 = \{-1, 2, 3, 0, 12\} \Rightarrow \mu_L^3 = 3.2 \\ S_H^3 = \{27, 31, 59, 61, 34\} \Rightarrow \mu_H^3 = 42.4 \end{cases}$$

$$T_3 = \frac{\mu_L^3 + \mu_H^3}{2} = \frac{3.2 + 42.4}{2} = 22.8$$

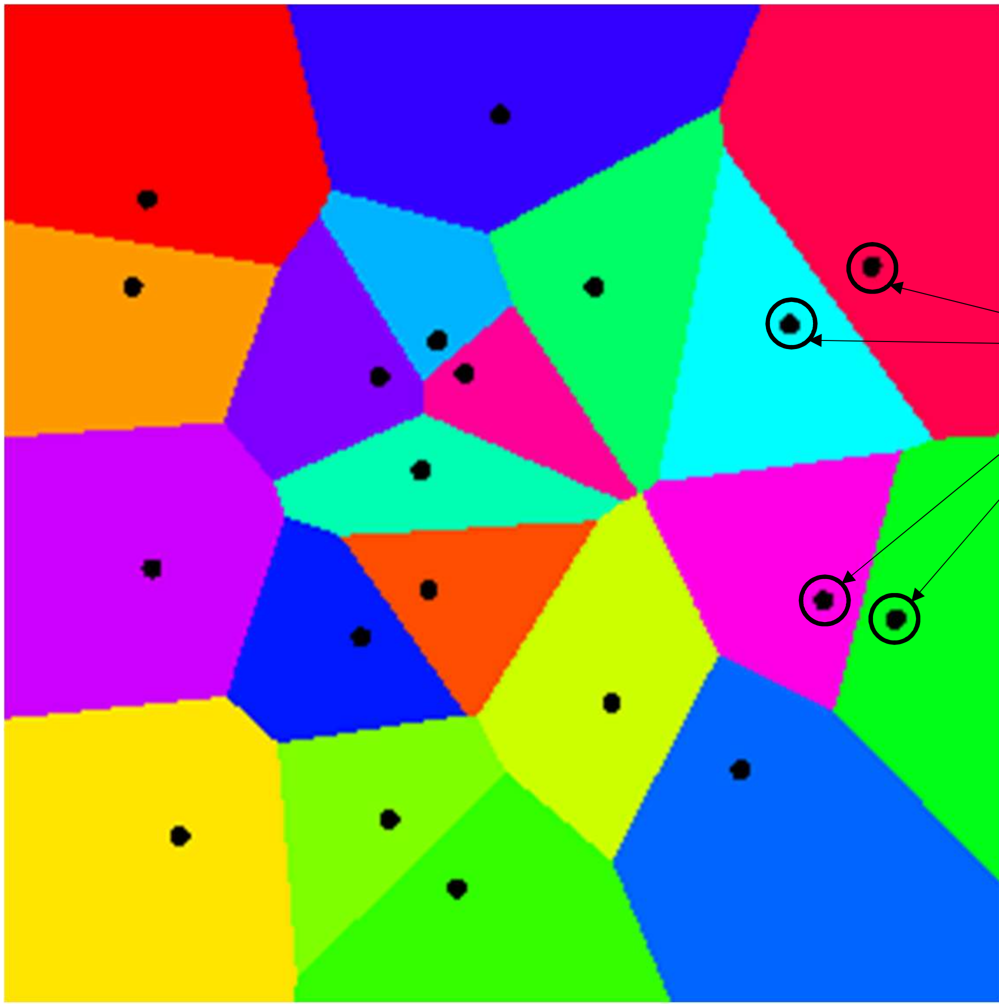
$$|T_3 - T_2| = 0.0$$

K-Means Clustering



K-Means Application: Image Segmentation

Voronoi Decomposition



A Point is Assigned to its Nearest Landmark Point

Landmark Points

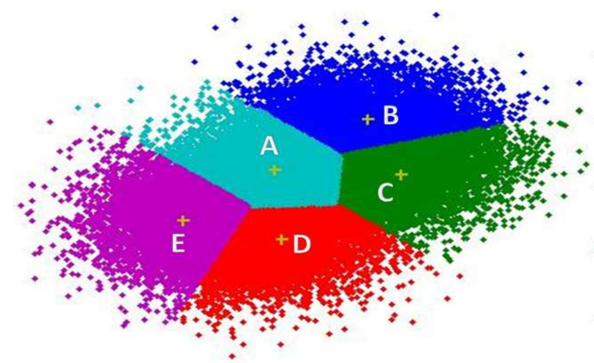
All Points assigned to Same Landmark Point have Similar Label

The Space is Decomposed into Partitions of Points having Similar Label

K-Means: Definitions

Input Dataset: $\boldsymbol{D} = \{\boldsymbol{x}_i \,; i = 1, \ldots n\}$

K-Means Clustering



$$\boldsymbol{D} = \bigcup_{j=1}^{K} \boldsymbol{D}_j \longrightarrow \left\{ \boldsymbol{\mu}_j = \frac{\sum_{\boldsymbol{x} \in \boldsymbol{D}_j} \boldsymbol{x}}{|\boldsymbol{D}_j|} \,; j = 1, \ldots K \right\}$$

K-Means: Initialization

$L_i^{(\tau)}$: Label of $\boldsymbol{x}_i$ at Iteration τ $(i = 1, \dots n)$

$\boldsymbol{\mu}_j^{(\tau)}$: Centroid of Cluster j at Iteration τ $(j = 1, \dots K)$

Randomly Initialize K-Means using Data: $\boldsymbol{\mu}_j^{(0)}; j = 1, \dots K$

OR

Randomly Initialize N Labels to K Values: $L_i^{(0)}; i = 1, \dots n$

K-Means: Algorithm

Iterations at Step τ, Labels $L_i^{(\tau-1)}; i = 1, \dots n$ are available

$$\boldsymbol{\mu}_j^{(\tau)} = \frac{\sum_{i=1}^{n} \boldsymbol{x}_i \delta\left[L_i^{(\tau-1)} - j\right]}{\sum_{i=1}^{n} \delta\left[L_i^{(\tau-1)} - j\right]}$$

Iteration: Centroid Re-estimation

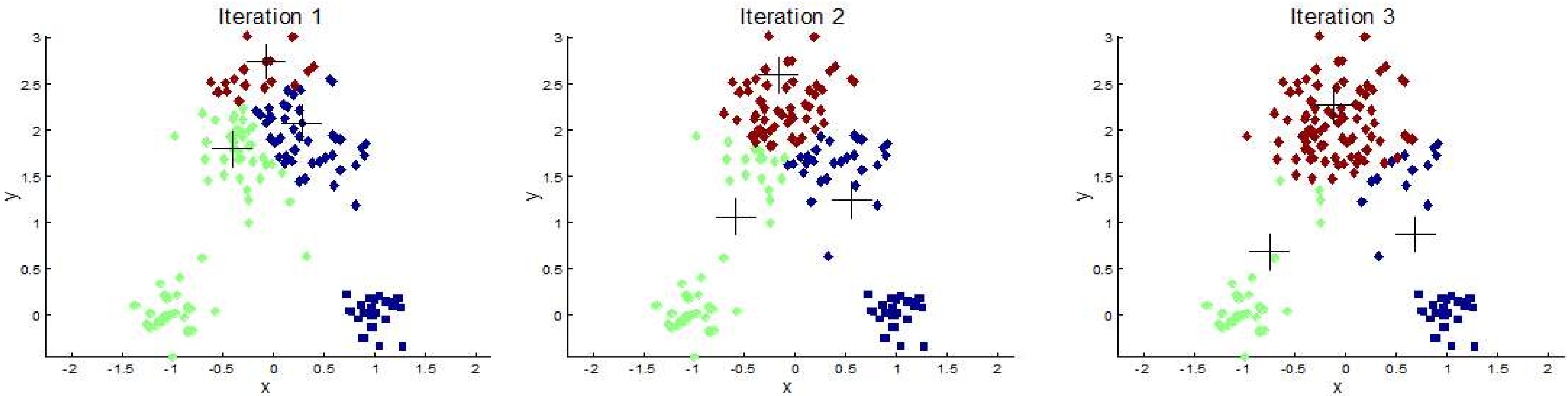
K-Means: Algorithm

Iteration: Label Update

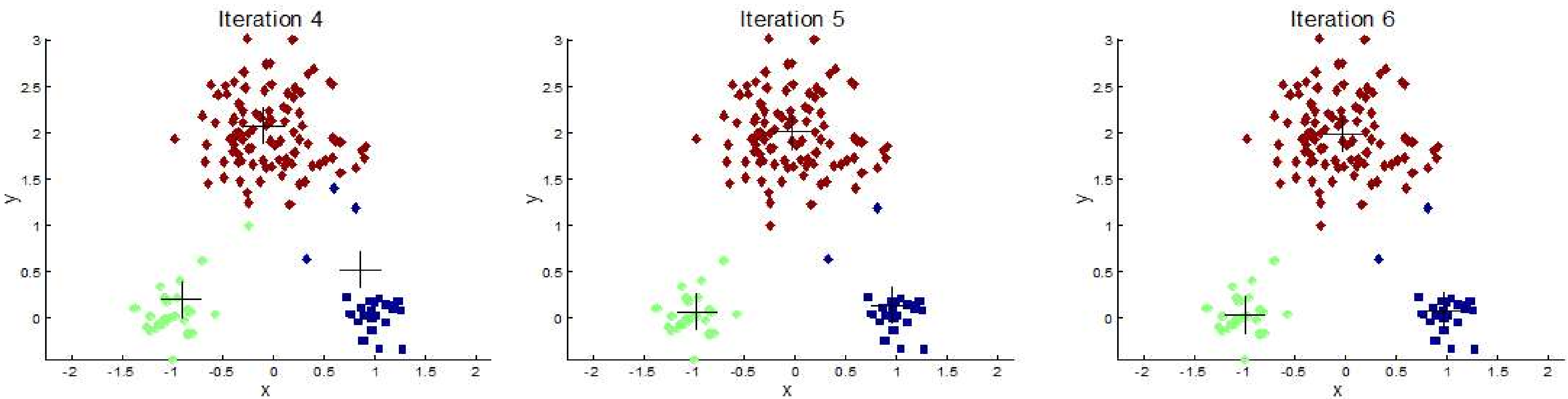
$$L_i^{(\tau)} = \underset{j=1,\ldots K}{\operatorname{argmin}} \left\| \boldsymbol{x}_i - \boldsymbol{\mu}_j^{(\tau)} \right\|_2^2$$

Termination Condition

$$\max_{j=1,\ldots K} \left\| \mu_j^{(\tau)} - \mu_j^{(\tau-1)} \right\| \leq \epsilon$$



K-Means: Iterative Centroid & Label Update



K-Means: Example

	DATA										MEANS		
#	**-1**	**27**	**31**	**2**	**59**	**3**	**61**	**34**	**0**	**12**	M_1	M_2	M_3
L_0	1	1	1	2	2	2	3	3	3	1			
L_1													
L_2													
L_3													
L_4													
L_5													
L_6													

DATA LABELS

K-Means: Example

#	-1	27	31	2	59	3	61	34	0	12	M_1	M_2	M_3
L_0	1	1	1	2	2	2	3	3	3	1			
L_1	1	3	3	1	3	1	3	3	1	1	17.25	21.33	31.67
L_2													
L_3													
L_4													
L_5													
L_6													

K-Means: Example

#	-1	27	31	2	59	3	61	34	0	12	M_1	M_2	M_3
L_0	1	1	1	2	2	2	3	3	3	1			
L_1	1	3	3	1	3	1	3	3	1	1	17.25	21.33	31.67
L_2	1	2	2	1	3	1	3	3	1	1	3.2	21.33	42.4
L_3													
L_4													
L_5													
L_6													

K-Means: Example

#	-1	27	31	2	59	3	61	34	0	12	M_1	M_2	M_3
L_0	1	1	1	2	2	2	3	3	3	1			
L_1	1	3	3	1	3	1	3	3	1	1	17.25	21.33	31.67
L_2	1	2	2	1	3	1	3	3	1	1	3.2	21.33	42.4
L_3	1	2	2	1	3	1	3	2	1	1	3.2	29	51.33
L_4													
L_5													
L_6													

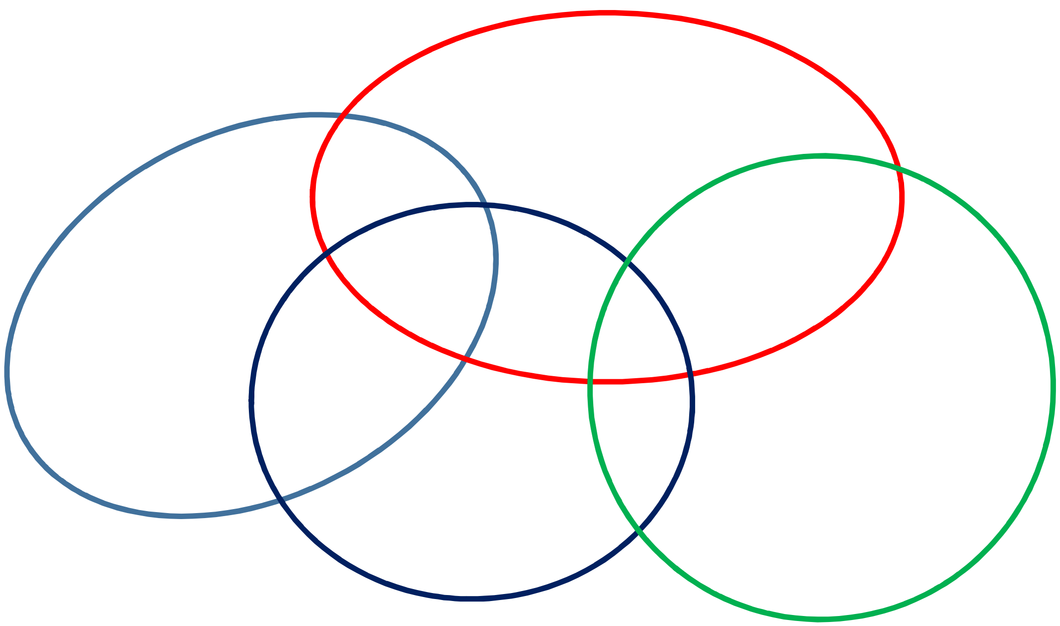
K-Means: Example

#	-1	27	31	2	59	3	61	34	0	12	M_1	M_2	M_3
L_0	1	1	1	2	2	2	3	3	3	1			
L_1	1	3	3	1	3	1	3	3	1	1	17.25	21.33	31.67
L_2	1	2	2	1	3	1	3	3	1	1	3.2	21.33	42.4
L_3	1	2	2	1	3	1	3	2	1	1	3.2	29	51.33
L_4	1	2	2	1	3	1	3	2	1	1	3.2	30.67	60
L_5													
L_6													

K-Means: Example

#	-1	27	31	2	59	3	61	34	0	12	M_1	M_2	M_3
L_0	1	1	1	2	2	2	3	3	3	1			
L_1	1	3	3	1	3	1	3	3	1	1	17.25	21.33	31.67
L_2	1	2	2	1	3	1	3	3	1	1	3.2	21.33	42.4
L_3	1	2	2	1	3	1	3	2	1	1	3.2	29	51.33
L_4	1	2	2	1	3	1	3	2	1	1	3.2	30.67	60
L_5	1	2	2	1	3	1	3	2	1	1	3.2	30.67	60
L_6											**Convergence**		

Membership Weighted K-Means



Membership Function

Parameters in Iteration τ
for cluster $\boldsymbol{\omega_j}$ $(j = 1, \dots K)$

$$\boldsymbol{x}_i \rightarrow P(\boldsymbol{\omega}_j | \boldsymbol{x}_{\boldsymbol{i}}, \boldsymbol{\theta}_j^{(\tau)})$$

Membership function
of data $\boldsymbol{x}_{\boldsymbol{i}}$ $(i = 1, \dots n)$

Membership Function

$$P\left(\boldsymbol{\omega}_j | \boldsymbol{x}, \boldsymbol{\theta}_j^{(\tau)}\right) = \frac{P\left(\boldsymbol{x} | \boldsymbol{\omega}_j, \boldsymbol{\theta}_j^{(\tau)}\right) P^{(\tau)}(\omega_j)}{P(\boldsymbol{x})}$$

$$\sum_{j=1}^{K} P\left(\boldsymbol{\omega}_j | \boldsymbol{x}, \boldsymbol{\theta}_j^{(\tau)}\right) = 1$$

$$P(\boldsymbol{x}) = \sum_{l=1}^{K} P\left(\boldsymbol{x} | \boldsymbol{\omega}_l, \boldsymbol{\theta}_l^{(\tau)}\right) P^{(\tau)}(\boldsymbol{\omega}_l)$$

Cluster Parameters

Cluster Mean

Cluster Prior

$$\boldsymbol{\theta}_j^{(\tau)} = \left\{\boldsymbol{\mu}_j^{(\tau)}, \boldsymbol{C}_j^{(\tau)}\right\}; P^{(\tau)}(\omega_j)$$

Cluster Covariance

Cluster Parameter Update: Mean

$$\boldsymbol{\mu}_j^{(\tau)} = \frac{\sum_{i=1}^{n} \boldsymbol{x}_i P\left(\omega_j | \boldsymbol{x}_i, \boldsymbol{\theta}_j^{(\tau-1)}\right)}{\sum_{i=1}^{n} P\left(\omega_j | \boldsymbol{x}_i, \boldsymbol{\theta}_j^{(\tau-1)}\right)}$$

K-Means: $\boldsymbol{\mu}_j^{(\tau)} = \frac{\sum_{i=1}^{n} \boldsymbol{x}_i \delta\left[L_i^{(\tau-1)} - j\right]}{\sum_{i=1}^{n} \delta\left[L_i^{(\tau-1)} - j\right]}$

Cluster Parameter Update: Covariance Matrix

$$\boldsymbol{Z}_{ij}^{(\tau-1)} = \left(\boldsymbol{x}_i - \boldsymbol{\mu}_j^{(\tau-1)}\right)\left(\boldsymbol{x}_i - \boldsymbol{\mu}_j^{(\tau-1)}\right)^T$$

$$\boldsymbol{C}_j^{(\tau)} = \frac{\sum_{i=1}^{n} P\left(\omega_j | \boldsymbol{x}_i, \boldsymbol{\theta}_j^{(\tau-1)}\right) \boldsymbol{Z}_{ij}^{(\tau-1)}}{\sum_{i=1}^{n} P\left(\omega_j | \boldsymbol{x}_i, \boldsymbol{\theta}_j^{(t)}\right)}$$

Cluster Parameter Update: Prior

$P(\omega_j)$: Proportion of Points in Cluster ω_j

$P(\omega_j|\boldsymbol{x}_i)$: Membership of $\boldsymbol{x}_i$ in Cluster ω_j

$$P^{(\tau)}(\omega_j) = \frac{\sum_{i=1}^{n} P\left(\omega_j|\boldsymbol{x}_i, \theta_j^{(\tau-1)}\right)}{n}$$

The Dataset

6	**-4**	**4**	**7**	**0**	**-7**	**1**	**-8**	**0**	**-7**
8	**-9**	**-4**	**-5**	**0**	**8**	**10**	**-1**	**6**	**7**

Membership Weighted K-Means: Initialization

6	**-4**	**4**	**7**	**0**	**-7**	**1**	**-8**	**0**	**-7**
8	**-9**	**-4**	**-5**	**0**	**8**	**10**	**-1**	**6**	**7**

$$\mu_1(0) = \begin{bmatrix} 1.5 \\ -7 \end{bmatrix} \qquad \mu_2(0) = \begin{bmatrix} -1.5 \\ 2 \end{bmatrix} \qquad \mu_3(0) = \begin{bmatrix} -3 \\ 8.5 \end{bmatrix}$$

$$C_1(0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad C_2(0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad C_3(0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\pi_1(0) = \frac{1}{3} \qquad \pi_2(0) = \frac{1}{3} \qquad \pi_3(0) = \frac{1}{3}$$

MW K-Means & K-Means

	$\begin{bmatrix}6\\8\end{bmatrix}$	$\begin{bmatrix}-4\\-9\end{bmatrix}$	$\begin{bmatrix}4\\-4\end{bmatrix}$	$\begin{bmatrix}7\\-5\end{bmatrix}$	$\begin{bmatrix}0\\0\end{bmatrix}$	$\begin{bmatrix}-7\\8\end{bmatrix}$	$\begin{bmatrix}1\\10\end{bmatrix}$	$\begin{bmatrix}-8\\-1\end{bmatrix}$	$\begin{bmatrix}0\\6\end{bmatrix}$	$\begin{bmatrix}-7\\7\end{bmatrix}$
$P(x\|\omega_1)$										
$P(x\|\omega_2)$										
$P(x\|\omega_3)$										

$$P(x|\omega_j) = \frac{1}{(2\pi)^{\frac{d}{2}}\left|C_j\right|^{\frac{1}{2}}} exp\left\{-\frac{1}{2}\left(x-\mu_j\right)^T C_j^{-1}\left(x-\mu_j\right)\right\}$$

MW K-Means & K-Means

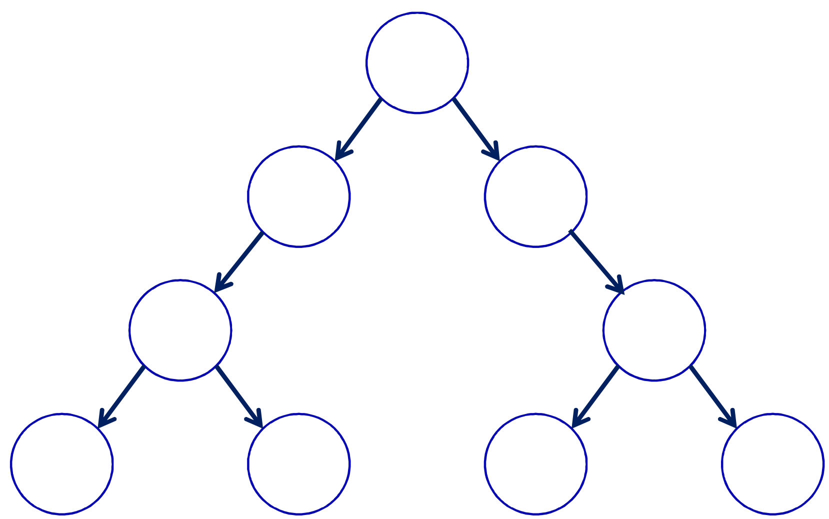
	$\begin{bmatrix}6\\8\end{bmatrix}$	$\begin{bmatrix}-4\\-9\end{bmatrix}$	$\begin{bmatrix}4\\-4\end{bmatrix}$	$\begin{bmatrix}7\\-5\end{bmatrix}$	$\begin{bmatrix}0\\0\end{bmatrix}$	$\begin{bmatrix}-7\\8\end{bmatrix}$	$\begin{bmatrix}1\\10\end{bmatrix}$	$\begin{bmatrix}-8\\-1\end{bmatrix}$	$\begin{bmatrix}0\\6\end{bmatrix}$	$\begin{bmatrix}-7\\7\end{bmatrix}$
$P(\omega_1\|x)$										
$P(\omega_2\|x)$										
$P(\omega_3\|x)$										

$$P(\omega_j|\boldsymbol{x}) = \frac{P(\boldsymbol{x}|\omega_j)P(\omega_j)}{P(\boldsymbol{x}|\omega_1)P(\omega_1) + P(\boldsymbol{x}|\omega_2)P(\omega_2) + P(\boldsymbol{x}|\omega_3)P(\omega_3)}$$

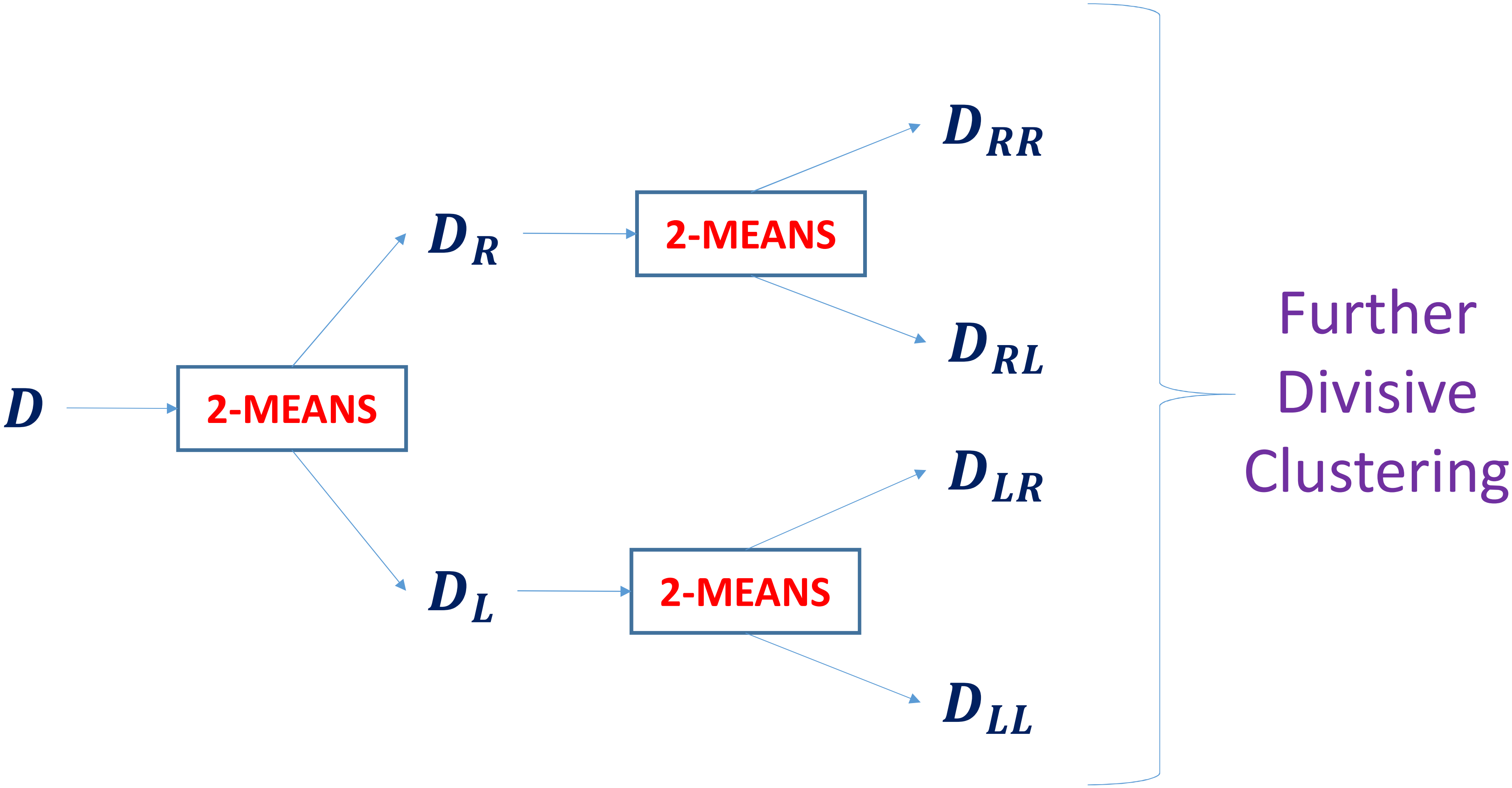
MW K-Means & K-Means

	$P(\omega_1 \vert x)$	…	$P(\omega_j \vert x)$	…	$P(\omega_K \vert x)$
x_1			Adds Up To ONE along Rows		
$\vdots$					
x_i			$P(\omega_j \vert x_i)$		
$\vdots$					
x_n			Has Binary Values for K-Means		

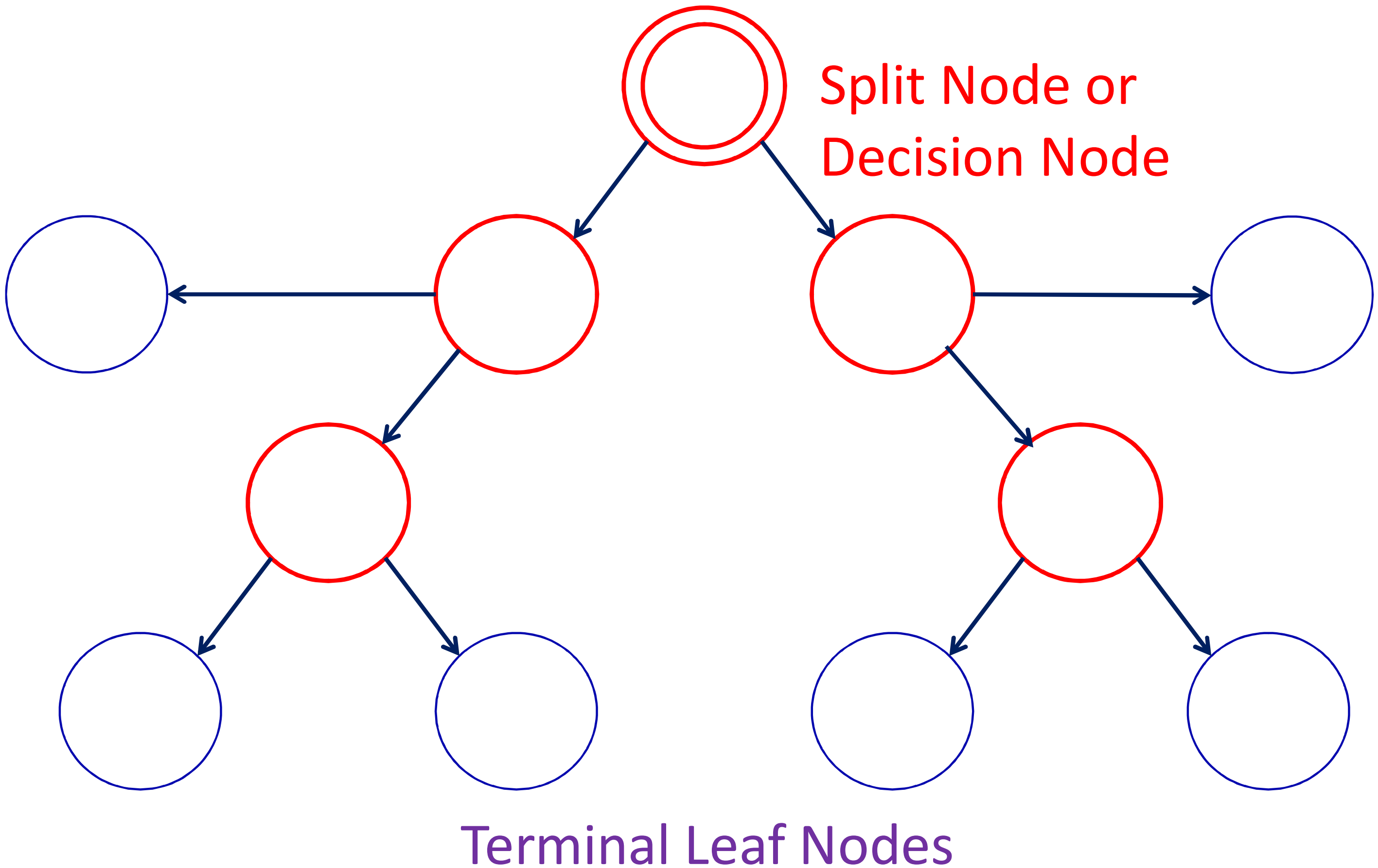
Hierarchical K-Means



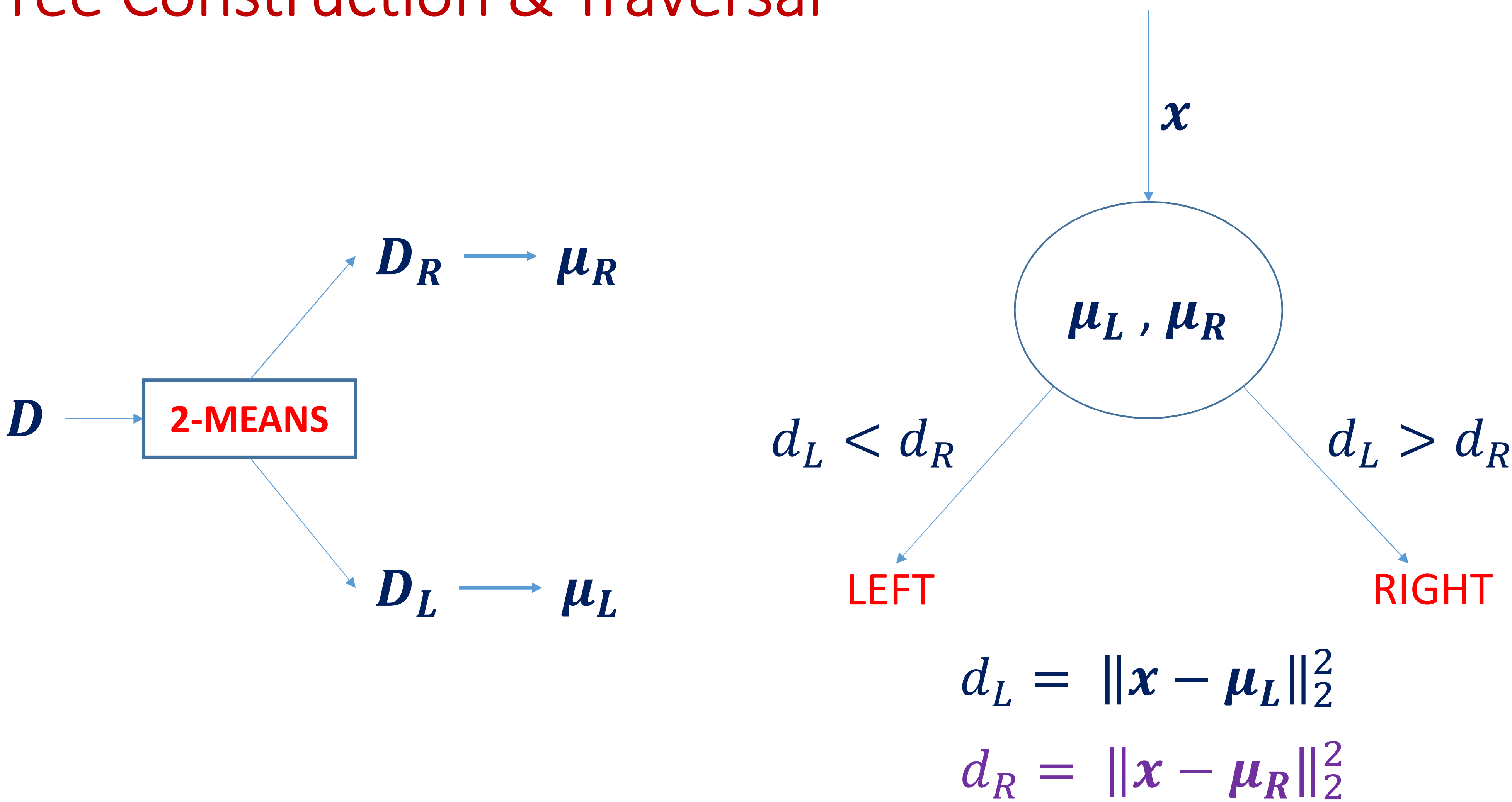
Divisive Clustering



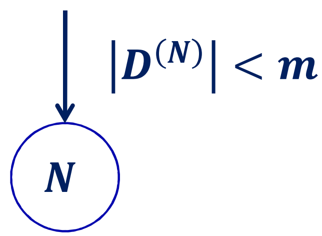
Nodes In The Tree



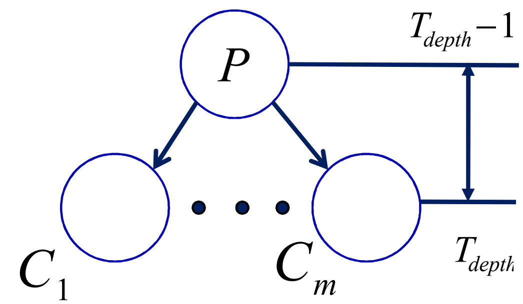
Tree Construction & Traversal



Termination Criteria



Insufficient Input to Node



Nodes at Maximum Allowable Tree Depth

$$checkSplitConditions\left(\boldsymbol{D}^{(N)}\right)$$

Hierarchical K-Means Tree Construction

1. Input to Node $\boldsymbol{N}$ is $\boldsymbol{D}^{(\boldsymbol{N})}$
2. Evaluate $splitFlag = checkSplitConditions\left(\boldsymbol{D}^{(\boldsymbol{N})}\right)$
3. IF $splitFlag = FALSE$ THEN $\boldsymbol{N} \rightarrow LeafNode$ ELSE
4. Clustering: $\left\{\boldsymbol{\mu}_{\boldsymbol{L}}^{(\boldsymbol{N})}, \boldsymbol{\mu}_{\boldsymbol{R}}^{(\boldsymbol{N})}\right\} = twoMeansClustering\left(\boldsymbol{D}^{(\boldsymbol{N})}\right)$
5. Compute: $\boldsymbol{d}_{\boldsymbol{L}}^{(\boldsymbol{N})}(\boldsymbol{x}) = \left\|\boldsymbol{x} - \boldsymbol{\mu}_{\boldsymbol{L}}^{(\boldsymbol{N})}\right\|_2^2$ and $\boldsymbol{d}_{\boldsymbol{R}}^{(\boldsymbol{N})}(\boldsymbol{x}) = \left\|\boldsymbol{x} - \boldsymbol{\mu}_{\boldsymbol{R}}^{(\boldsymbol{N})}\right\|_2^2$; $\forall \boldsymbol{x} \in \boldsymbol{D}^{(N)}$
6. Split: IF $\boldsymbol{d}_{\boldsymbol{L}}^{(\boldsymbol{N})}(\boldsymbol{x}) < \boldsymbol{d}_{\boldsymbol{R}}^{(\boldsymbol{N})}(\boldsymbol{x})$ PUSH $\boldsymbol{x} \rightarrow \boldsymbol{D}_L^{(N)}$; ELSE PUSH $\boldsymbol{x} \rightarrow \boldsymbol{D}_R^{(N)}$
7. Check: $\left[\boldsymbol{D}^{(\boldsymbol{N})} = \boldsymbol{D}_L^{(N)} \cup \boldsymbol{D}_R^{(N)}\right] \wedge \left[\boldsymbol{D}_L^{(N)} \cap \boldsymbol{D}_R^{(N)} = \varphi\right]$
8. Children Node Inputs: $\boldsymbol{D}_L^{(N)} \rightarrow N_L$ AND $\boldsymbol{D}_R^{(N)} \rightarrow N_R$
9. Splitting is Performed Recursively until Termination

The Dataset

6	**-4**	**4**	**7**	**0**	**-7**	**1**	**-8**	**0**	**-7**
8	**-9**	**-4**	**-5**	**0**	**8**	**10**	**-1**	**6**	**7**

Assignment: Perform Hierarchical K-Means Clustering to obtain 4 Clusters

K-Means: Advantages

- Simple Algorithm & Easy to Implement
- Easily Partitions Input Space
- Can be Parallelized for Large Data sets
- Produces Convex Clusters
- Can be used as a Preprocessing Stage
- Useful for Feature Learning as Well

K-Means: Drawbacks

- Very Slow for Big Data and Low Resource
- Not Suitable for Non-Convex Datasets
- Sensitive Towards Outliers
- Spatial Density Variation Affects Performance
- Number of Clusters need to be Specified

Thank You